

Robot car test

Basic test to verify the robot car's key mounted components and STM32F407VET6 30D.

- a) STM32F407VET6 30D inclusive of ICM-20948 IMU sensor
- b) Hall Encoder Motors
- c) Servo motor

Procedures:

- 1) Connect a USB-C cable from the STM32 board COM port (next to the blue potentiometer) to the computer.
- 2) Open the software application FlyMCU ver02188 and prepare to flash the Stm32_C30D.hex file to the controller board.

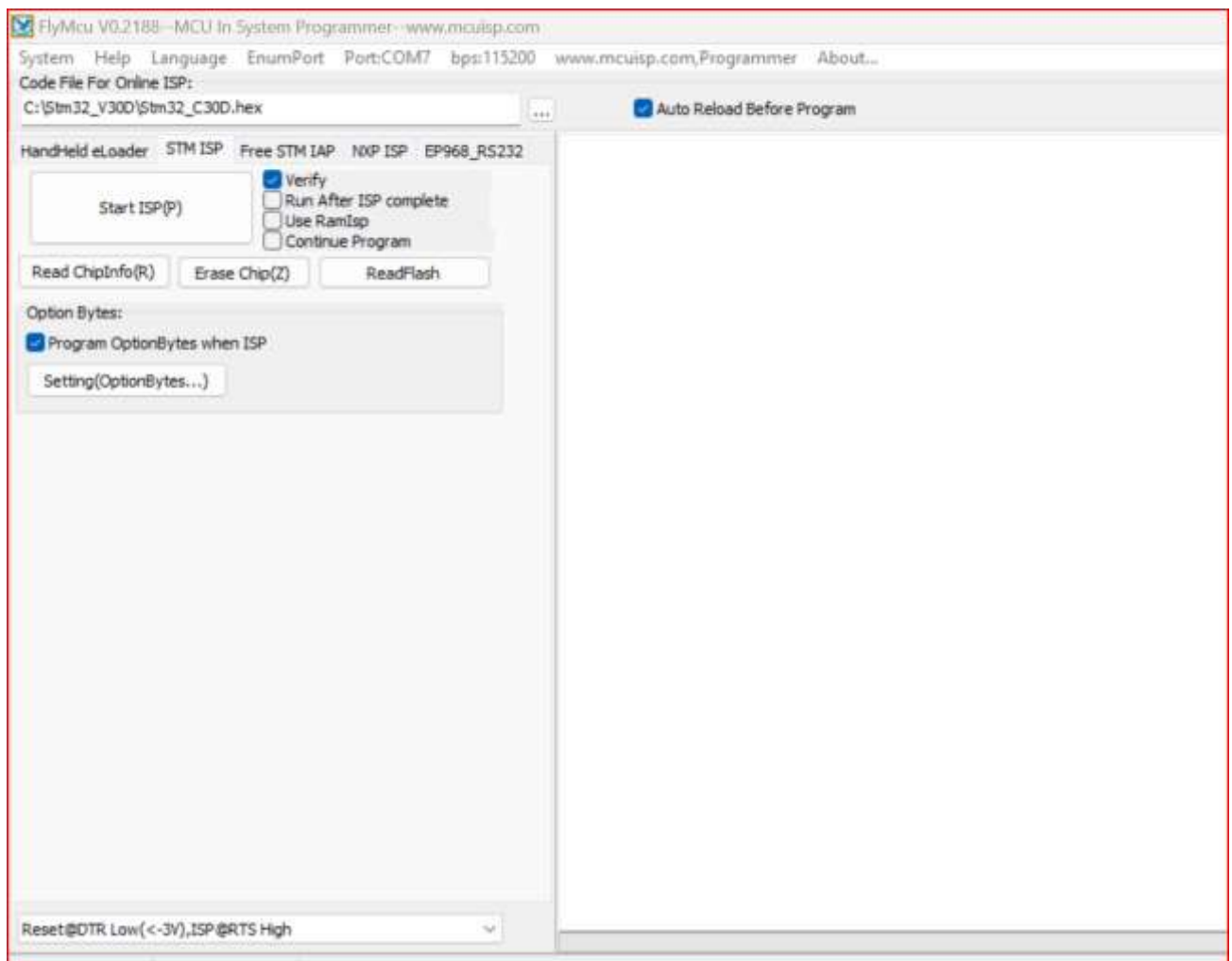


Fig. 1

- 1) Config the setting "Reset@DTR Low (<-3V), ISP @RTS High)" at the bottom of panel. See Fig.1.
 - 2) From "EnumPort" select the Com port related to Free Silicon Labs CP210x USB to UART Bridge and Set bps to 115200.
 - 3) Under "Code File For Online ISP" select the Stm32_C30D.hex file.
 - 4) Press the "Start ISP(P)" and the program will execute as seen in the right Window panel. See Fig
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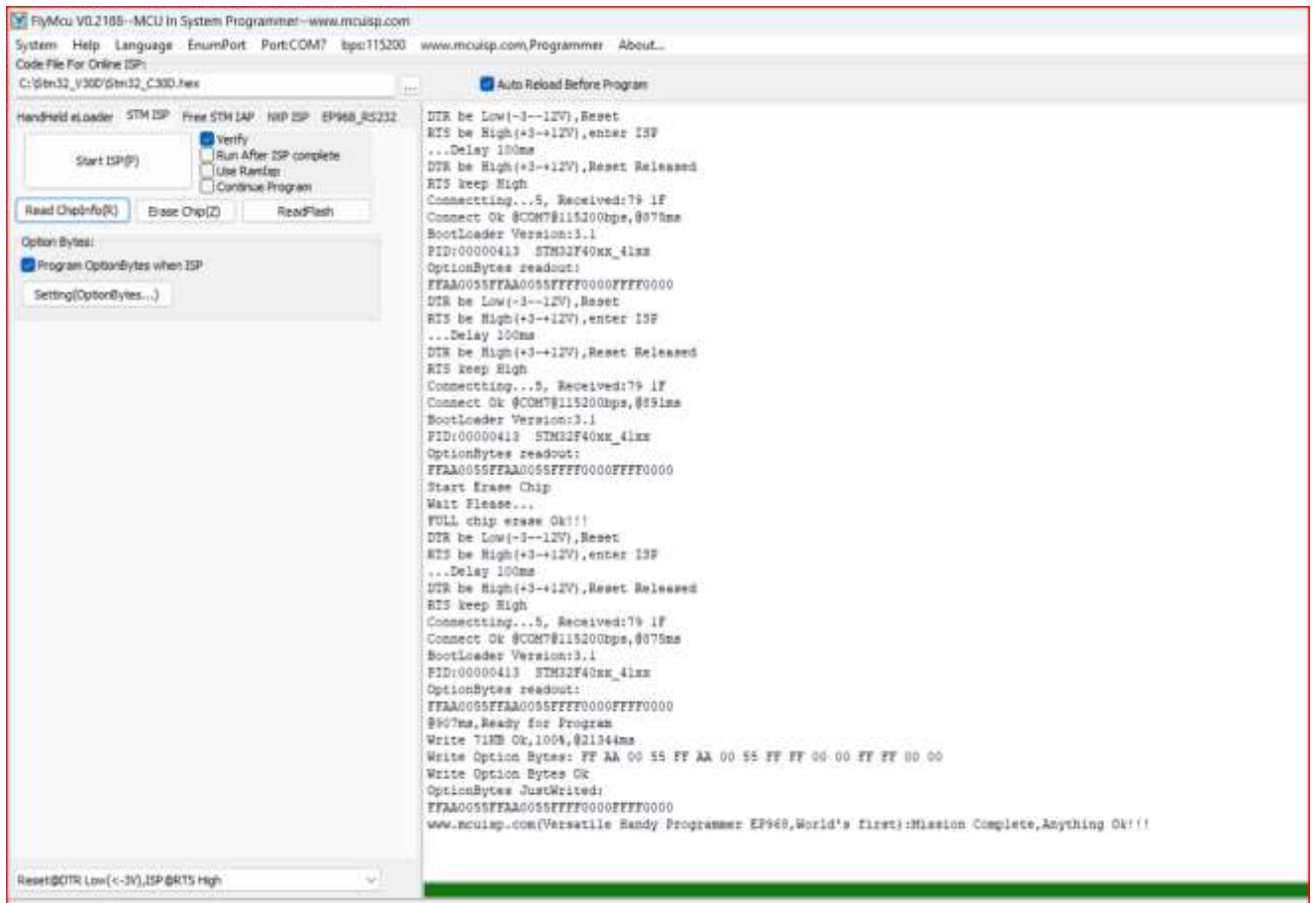


Fig 2.

- 5) Once the download is successful, close the application and prepare to test the robot car.
- 6) Connect the 3-pin servo motor cable to the header as shown in Fig 3.

Brown – Gnd
Red – 5V
Pin B15 – Signal



Fig 3

7. Connect motor cables to headers labelled Motor A and Motor B of the STM32 board, respectively.
8. Connect the **Right-angle DC cable with Red/Black leads** from the Output port of the battery pack to the blue terminal.
9. Tune the blue potentiometer to select “AKM” which displays on the OLED panel.

Press the “**user**” button for each step of the test execution.

1. Simultaneously switch on the STM32 board and press the “user” button.
2. Wait for blue LED3 to blink (about 10 sec).
- 3 Press the “user” button once for the OLED panel to display 1/8 (which means step 1).
4. Press again for the OLED panel to display 2/8.
5. Press again to execute the **Left and Right motors test**.

Observation

- a. Take note of the motor A and B speeds displayed on the OLED panel. The motor speed readings should reflect equivalent values.
 - b. The rear wheels should rotate forward, then backward, and finally come to a stop.
6. Next, press the “user” button once for the OLED panel display 3/8.
 7. Press again to execute the **accelerometer and gyroscope test** (ICM-20948 IMU sensor).
 8. Move and tilt the STM32 controller board and observe the readings displayed on the OLED panel.

Observation

- a. The accelerometer reading on the OLED panel should stay between 16300 and 16400.
 - b. The gyroscope reading on the OLED panel should display a single digit.
9. Next, press the “user” button once for the OLED panel display 4/8.
 10. Press again to execute the **servo motor test**.

Observation

The front wheels turn to position left, followed by right, and then stop at the left position.

11. Double-press the “user” button to restore the front wheels to their original alignment.

View the MP4 video – Robot Car V30D test.MP4 attached.